Republic of the Philippines
EULOGIO "AMANG" RODRIGUEZ
INSTITUTE OF SCIENCE AND TECHNOLOGY
CAVITE CAMPUS
General Mariano Alvarez, Cavite

Monte Cristo Research and Educational Institute GMA

Cavite School Website with Online Enrollment System

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by

JONELYN M. ANOS

RALLION MIGUEL ALFONSO S. HOSEÑA

CRISJOHN D. MERCADO

REICER O. RAMBONANZA

JOSHUA E. TOGORES

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CHAPTER 1

The Problem and Its Background

Introduction

The rapid advancement of modern technology has significantly evolved the way people handle information, conduct transactions, and give services to various fields. Among these advancements, the internet has become a powerful tool, that is applied in e-commerce, food ordering platforms, enrollment systems, and more. In today's society, almost everyone relies on technology to accomplish tasks conveniently, as it provides a vital role in handling a wide range of everyday activities (RM Bracero et al., 2025). Despite this progress, many schools in the Philippines still rely on manual enrollment procedures. This traditional method often leads to delays, long queues, disorganized documents, and inaccurate data (Campos, 2022). While some schools prefer to stick with these usual processes, others are open to adopting digital solutions but lack the necessary resources for implementation. This situation shows the growing need for an online enrollment platform that is user-friendly, reliable, and efficient especially

as student populations continue to increase. With such a platform, schools, parents, and students can benefit from a more convenient and time-saving process. As enrollment numbers rise, there is a clear need for a more organized system that combines both in-person and online process. A digital solution can improve accessibility and allow enrollment procedures to serve more individuals efficiently, aligning with the evolving standards of modern education systems (Bognot, 2021).

Theoretical Framework

Combining educational and technological theories, the online enrollment system for a private school in General Mariano Alvarez, Cavite, was developed with the goal of making the process easy, effective, and fulfilling for both parents and school personnel. The use of simple digital tools that promote active participation and improved comprehension is fundamentally supported by constructivist learning theory, which makes the enrollment process more interesting and easy to use. The TPACK Framework makes sure that the system's technology is well matched with the educational objectives and administrative

Requirements of The Diffusion of Innovation Theory offers guidance on how to successfully and progressively introduce change among parents, teachers, and administrators to facilitate the school community's seamless transition to this new digital procedure. Systems Theory, on the other hand, improves data management, coordination, and communication while highlighting the system's significance as a component of the school's larger ecosystem. To make the enrollment process easier for users—especially parents who are not comfortable with digital platforms, Media Richness Theory recommends the use of interactive forms, clear images, and interesting material. Finally, by incorporating Filipino values, language, and customs into the system's architecture and content, Culturally Responsive Pedagogy reminds us to make sure it feels comfortable and familiar. When combined, these theories provide a strong basis for a digital platform that is not only useful but also inclusive, interesting, and genuinely supportive of the requirements of the local school community.

Conceptual Framework

Figure 1: Conceptual Framework of the Study

INPUT	PROCESS	OUTPUT
1. The issues in the existing system as to: 1.1 Functional Completeness; 1.2 Performance Efficiency; 1.3 Compatibility; 1.4 Usability; 1.5 Reliability; 1.6 Security; 1.7 Maintainability; 1.8 Portability; 2. Based on the data gathered, what website may be developed? 3. The evaluators assess the developed website as to: 3.1 Functional Completeness; 3.2 Performance Efficiency; 3.3 Compatibility; 3.4 Usability; 3.5 Reliability; 3.6 Security; 3.7 Maintainability; 3.8 Portability; 4. From the findings, what implementation program may be proposed?	Agile Methodology • Requirement Gathering and Analysis • System Design • Implementation • Integration • Testing • Deployment • Maintenance Analysis and interpretation of gathered data using statistical treatment	MONTE CRISTO RESEARCH AND EDUCATIONAL INSTITUTE GMA CAVITE SCHOOL WEBSITE WITH ONLINE ENROLLMENT SYSTEM

Feedback

Figure 1 presents the conceptual framework of the study using the Input-Process-Output (IPO) model. This framework illustrates how the study progresses from problem identification to the development of a functional online enrollment system that meets school needs.

The input stage consists of the identified issues in the existing enrollment process, focusing on areas such as functional completeness, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. These concerns serve as the foundation for determining what type of system needs to be developed. In addition, the input covers the gathered data and the evaluators' assessment of the system based on these same criteria. The findings from these inputs helped shape the direction of system development and provided a basis for proposing an appropriate implementation plan.

The process stage applies the Agile Methodology, which involves iterative and flexible development steps. This includes planning, requirement gathering, system design, implementation, integration, testing, deployment, and maintenance. The researchers also conducted analysis and

interpretation of the gathered data using statistical treatment to ensure that the features of the system address the problems identified during the input phase. This cyclical process ensured that improvements were incorporated throughout development, leading to a more responsive and functional system.

The output stage is the completed Online Enrollment System for Monte Cristo Research and Educational Institute-GMA, Cavite. The system is designed to streamline the enrollment workflow and address the issues identified in the input stage. It aims to provide secure access, improve efficiency, enhance user experience, and support administrative tasks. The final product reflects the goals of the study by offering a functional, user-friendly, and data-driven solution tailored to the school's needs.

Statement of the Problem

This study developed an Enrollment System for Monte Cristo Research and Educational Institute GMA Cavite, design implementation, and efficiency properties among the intended

users of the program.

Specifically, it aimed to answer the following question:

1. What are the issues existing in the system in the following aspects:

1.1 Functional Completeness;

1.2 Performance Efficiency;

1.3 Compatibility;

1.4 Usability;

1.5 Reliability;

1.6 Security;

1.7 Maintainability; and

1.8 Portability?

2. Based on the data gathered, what website may be developed?

3. How do evaluators assess the developed website as to:

3.1 Functional Completeness;

3.2 Performance Efficiency;

3.3 Compatibility;

3.4 Usability;

3.5 Reliability;

3.6 Security;

3.7 Maintainability; and

3.8 Portability?

4. From the findings, what implementation program may be proposed?

Scope and Limitation of the Study

The scope of the study includes the design and development of a website with an integrated enrollment system for Monte Cristo Research and Educational Institute GMA Cavite. The study focuses on enhancing user interface design, system functionality, data security, and accessibility for student, parents, teachers and school administrators. The study is limited to the enrollment process only and does not include other school management systems such as grading, attendance tracking, and school portal. Other features outside of the enrollment process will not be covered by this study.

Significance of the Study

This study aims to develop and implement a functional

website with an enrollment system for Monte Cristo Research and Educational Institute GMA Cavite for an easier school administration, enhancement in enrollment procedures, communication with parents & the community including other scholastic metrics. This study will be useful to the following stakeholders:

Students They will gain access to a centralized online portal, streamlining their access to school-related data and resources.

Parents and Guardians They will benefit from a more efficient and convenient online enrollment process, as well as simplified access to school activities, updates on student progress, important school notices, and direct communication channels with teachers.

Teachers and Staff This system promises to streamline enrollment, thereby reducing manual labor and improving the methods for managing data.

School Administration They will acquire a modern instrument for managing enrollment, effectively communicating, and enhancing the school's operational efficiency and public image.

Future Researchers The study will serve as a valuable case study and provide insights into the development and deployment of integrated school websites and enrollment systems within a Philippine private school setting.

Definition of Terms

For better understanding and interpretation of this study, the following terms are operationally defined:

Enrollment System A systematic procedure, often facilitated by a software application, utilized by educational institutions to manage the processes of student admission and registration.

Integrated System A system where various components or modules function in unison, sharing data and functionality. In this context, it refers to the combination of the website and the enrollment system into a single platform.

Online Platform A web-based environment or system accessible via the internet, offering a spectrum of services and information.

School Administration The individuals or department accountable for overseeing the operational aspects of the

school.

Streamlined Process A process that has been made more efficient and less complex, resulting in fewer unnecessary steps and reduced workload.

Stakeholders Individuals or groups with an interest in or affected by the project or study outcomes. This includes students, parents and guardians, teachers and staff, school administration, and future researchers.

CHAPTER 2

Review of Related Literature and Studies

Local Literature

Estanislao et al. (2020) piloted an E-Learning Outreach Program for public elementary schools, providing insights into the basic infrastructure and digital literacy required for successful e-learning platform deployment. The findings highlighted how the provision of internet-connected e-learning stations significantly improved students' digital competence and readiness for web-based education systems. This suggests that setting up a school website could also serve as a foundation for broader e-learning initiatives.

De Guzman and Cruz (2019) emphasized the vital role of websites in improving communication between the school and parents. The findings showed that elementary schools which adopted digital platforms like websites experienced more engaged and informed parents. The authors noted that websites allowed parents to monitor school activities, events, and even student progress despite their busy schedules. Furthermore, the study suggested that digital communication platforms foster a more collaborative

relationship between schools and families.

Mateo (2021) investigated the results and revealed that digital literacy among teachers plays a crucial role in the maintenance and effectiveness of school websites. Mateo found that when teachers were trained in basic website management, they became more proactive in updating content and ensuring the website's relevance. This study implies that capacity-building programs focused on ICT skills are vital for the successful implementation and sustainability of school websites.

Villanueva (2017) discussed how the integration of ICT programs helped in launching school websites. She emphasized the supportive role of DepEd's ICT initiatives, which encouraged schools to digitize communication and administrative services. Villanueva observed that schools with active websites also enhanced their teaching strategies by providing online learning resources and announcements, thus promoting a more modern educational environment even at the elementary level.

Garcia (2020) said that apart from reading, writing, and arithmetic, media literacy has become one of the necessary literacy that the children should acquire for

them to successfully navigate their ever-changing digital media environments. However, an equally heavy work is needed to advance media literacy advocacy among our younger media consumers-the Philippine basic education curriculum has yet to fully integrate media literacy for elementary and junior high school learners, and we have yet to see more culturally sensitive, child-friendly, and age-appropriate educational materials that focus on enhancing one's media literacy skills and competencies.

Foreign Literature

According to Zen et al. (2023), online registration of new students is a solution for more transparent, easy to collect data, and informative new student registration. Not many schools still implement or use information systems in the new student registration process.

Onyenwe et al. (2022) said that parents play vital roles in contributing to students' achievement. They need to be actively involved in monitoring their children's status and daily performance in school using web-based system.

Research conducted by Asrin et al. (2023) about

implementing a web-based school information system at SDN 14 Pontianak City, it reveals that school information systems address challenges related to ineffective internal information delivery within the school.

Limbong et al. (2021) pointed out that information systems can be an effective assistant in the enrollment process carried out by teachers or teaching staff as it lessens the manual works in the school.

Sara et al. (2022) stated that students' grades and information will be easier to manage with the help of a School website.

Local Studies

Based from the study conducted by Mina et al. (2021), the students marked the online enrollment system very good because it is better than the physical enrollment process. The online registration system is far better than the walk-in queues owing to the fact that new technology leads to a change in how people access different services.

According to Grepon et al. (2021), as the number of students grow bigger, it is in need of a systematic enrollment process where face to face and online process

should be taken into consideration. The enrollment process could be better and become readily accessible to the users. The procedure will then be conducted at optimum operation, catering more individuals at the same time with the new standard process of enrollment. The most common problems encountered by the school at the end of each semester are the student enrollment process, which typically lasts 2-3 weeks because most of the admission and storage of student information is done manually. Retrieval and generation of data and information is also a concern, as it takes a longer period to consolidate information that is required to complete the course.

Hayagan (2022) stated that Central Philippines State University San Carlos Campus Online Enrollment System was efficient, effective, user-friendly and easy to understand. The enrollment procedure is a huge help to the student, parent and to the school. The students of CPSU San Carlos Campus easily understands the content and meaning of each enrollment procedures implemented online. The students also agreed that the online enrollment system saves time effectively. The school provides wide access and stores accurate student accounts with the help of the

enrollment methods implemented online. Through the help of an online enrollment system, the school can easily track and monitor the needs of the school and the student. The implementation is highly suggested by the respondents to go on due to the efficiency of the said online system.

Caratiquit (2021) stated that a school system or website might be used by academic institutions such as colleges and universities to keep student and staff records easily. Additional benefits include the ability to save money, time, and effort by using a web-based information and publication system for schools. If an institution has a well-designed website, it will be possible to create online portal brochures. The school's management will not have to worry about running print or television advertisements to promote the institution. Additionally, the necessity that parents and other members of the school community receive printed documents and announcements is gone.

Based from the study conducted by Bracero et al. (2025), the State Universities and Colleges continuously working on the implementation of information systems particularly the enrolment system. This effort among

educational institutions to automate enrolment system integrates various processes into an information system which consolidates data into a database server, thus, access to data and information becomes more efficient. The implementation of such systems also lessens time requirement for a certain transaction to finish and therefore increases personnel efficiency and productivity. Enrollment system is highly usable, secured, efficient, and provides a fast and easy way to manage enrollee records, updating students/enrollee data, viewing the total number of enrolled students, adding of courses, subjects, and class schedules.

Foreign Studies

According to Zen et al. (2023), online registration of new students is a solution for more transparent, easy to collect data, and informative new student registration. Not many schools still implement or use information systems in the new student registration process.

Onyenwe et al. (2022) said that parents play vital roles in contributing to students' achievement. They need to be actively involved in monitoring their children's

status and daily performance in school using web-based system.

Research conducted by Asrin et al. (2023) about implementing a web-based school information system at SDN 14 Pontianak City, it reveals that school information systems address challenges related to ineffective internal information delivery within the school.

A cloud-based student enrollment system for primary schools in Taiwan that utilized Node.js, MongoDB, and cloud hosting through AWS by Chen and Li (2021) revealed that a marked improvement in enrollment accuracy and efficiency while minimizing paper usage. This approach reflects modern cloud practices relevant to the current research, especially in the areas of scalability and data integrity.

Garcia and Torres (2022) who implemented a multilingual school admission platform in Spain to accommodate the country's diverse linguistic demographics that was built with Django and PostgreSQL, the system allowed applicants to complete the process in their native language, improving user experience and accessibility. Their research showed that localized interfaces boosted

application completion rates and user satisfaction principles that align with the user-centered design aspect of the Monte Cristo Research and Educational Institute GMA Cavite website project.

Synthesis

Modern technology significantly alleviates the challenges associated with manual tasks and processes across various fields, including the educational sector. In school settings, the integration of technology streamlines essential services such as enrollment and information dissemination. When developing a functional, efficient, and well-structured system or website, it is crucial to prioritize the needs and experience of the users.

This review of related literature and studies demonstrates that a school website, integrated with an enrollment system offers significant benefits to students, parents, teachers, and school administrators by effectively facilitating the dissemination of information and streamlining enrollment processes through reduced manual intervention.

CHAPTER 3

Methodology

This chapter presents the research design, the respondents and the sampling technique, the research instruments used in gathering data, the data gathering procedures and the statistical treatments of data.

Research Design

In developing the online enrollment system, the researchers applied a combination of descriptive and developmental research approaches to obtain both qualitative and quantitative data. The descriptive aspect focused on identifying the challenges in the current enrollment process by using a tool such as survey. Through this, the researchers were able to outline the specific features and improvements needed in the system.

The developmental approach, on the other hand, was used to guide the actual development and enhancement of the system. It supported the design and development, ensuring that the platform responds to the needs of students, parents, and school administrators. The system's quality was evaluated

using the ISO 25010 standards, covering areas like usability, functionality, performance, security, reliability, compatibility, portability, and maintainability. By integrating these considerations, the online enrollment system was built not only to function effectively but also to resolve the issues found in the school's previous enrollment process.

Population and Sample

Table 1

Sample Coverage of the Respondents in the Study

Evaluators	Population (N)	Sample (n)	Percentage (%)
IT/CS Students	347	30	8.6%
IT Professionals	20	10	50%
Total	367	40	10.9%

The table presents the population and sample distribution of the respondents who evaluated the online enrollment system. IT/CS students make up the largest group,

with 30 participants representing 8.6% of their total population. Their inclusion ensures feedback from frequent system users who can assess usability and functionality from a student perspective. Meanwhile, IT professionals account for 10 respondents, comprising 50% of their population. Their participation provides technical insight into system performance, security, and reliability. The combination of students and expert evaluation supports a well-rounded and credible assessment of the system.

To ensure that feedback was gathered from individuals with experience and technical knowledge in evaluating a system, specifically an online enrollment system, the researchers employed purposive sampling. This non-probability sampling technique allowed the selection of respondents based on their roles and familiarity with the system evaluation, making them suitable sources of input for the evaluation and improvement of the proposed system.

Respondents of the Study

The study involved forty (40) respondents from the two (2) key groups: IT/CS Students and IT Professionals who are

qualified to assess the online enrollment system based on ISO 25010 software quality characteristics.

Research Instrument

The main instrument used in the study was a Google Forms-based evaluation tool designed to collect the data needed to assess the online enrollment system. The form was developed in alignment with the ISO 25010 software quality characteristics to ensure that the system is evaluated based on recognized industry standards.

The Google Form began with a brief introduction explaining the purpose of the study, along with a consent statement to inform respondents about confidentiality and the voluntary nature of their participation. This was followed by a short information section where respondents were asked to provide details such as their name, email, year and course, and evaluator type (IT/CS student or IT professional/faculty).

The evaluation section focused on key areas such as user satisfaction, system usability, performance, security, and overall effectiveness. Questions were patterned after the ISO 25010 characteristics, specifically functional completeness,

performance efficiency, compatibility, usability, reliability, security, portability, and maintainability. A Likert Scale format was used to gather measurable responses and determine the system's strengths and areas for improvement.

The respondents consisted of IT/CS students and IT professionals or faculty members, ensuring that the feedback came from individuals with relevant technical knowledge and familiarity with system evaluation. The data gathered from the instrument provided the basis for assessing the system's functionality, identifying gaps, and recommending enhancements.

Data Gathering Procedure

The data for this study were collected through a Google Forms survey designed specifically for the evaluation of the online enrollment system. The form was divided into three main sections. The first section gathered basic respondent information, including their name, email, year and course, and evaluator type (IT/CS student or IT professional/faculty). The second section focused on the assessment of the system based on ISO 25010 software quality characteristics,

particularly functionality, usability, reliability, maintainability, portability, accessibility, and security. The final section allowed respondents to provide comments and suggestions for further improvement of the system.

The link of the Google Form was sent to the evaluators with consent of the respondents to ensure voluntary participation and ethical data collection. Once responses were gathered, the data were organized, prepared, and reviewed to assess the system's overall effectiveness and to determine areas that may require enhancement. The collected feedback served as the basis for recommending improvements and finalizing the evaluation of the system.

Statistical Treatment of Data

The following statistical methods were applied to analyze the data collected from respondents:

1. Frequency and Percentage was utilized to determine the demographic distribution and responses of the participants regarding the system's usability, functionality, design, and content.

Formula:

$$\text{Percentage (\%)} = \left(\frac{f}{n}\right) x\, 100$$

f = frequency of each response

n = total number of respondents

2. Weighted mean was used to assess the respondents' level of satisfaction with the developed online enrollment system.

Formula:

$$\frac{\sum(f * x)}{n}$$

f = frequency of each scale point

x = corresponding scale weight (1 to 5)

n = total number of responses

Frequency – Refers to how many times a specific answer, value, or response appears in a set of data. It helps identify which responses are most or least common.

Percentage – Shows the portion or share of a particular

response compared to the total number of responses. It is expressed out of 100 to make comparisons clearer.

Weighted Mean – A type of average that gives more importance to certain responses based on their assigned weight or value. This is commonly used in Likert scale surveys to determine the overall level of agreement or evaluation.

For the Respondents:

Weighted Mean Range	Verbal Interpretation
4.20 - 5.00	Excellent
3.40 - 4.19	Very Good
2.60 - 3.39	Good
1.80 - 2.59	Fair
1.00 - 1.79	Poor

The five-point Likert Scale presents the interpretation scale used to analyze the weighted mean scores from the respondents. Each range corresponds to a descriptive rating that reflects the users' level of agreement or satisfaction with the features of the online enrollment system. Weighted mean scores from 4.20 to 5.00 are interpreted as Excellent,

indicating that the respondents strongly agree that the system performs very well in the evaluated aspects. Scores within 3.40 to 4.19 fall under Very Good, showing that most users are highly satisfied with the system's functionality and quality. A mean score of 2.60 to 3.39 is rated as Good, which suggests that the system meets expectations but may still have room for improvement. Scores ranging from 1.80 to 2.59 are classified as Fair, reflecting moderate satisfaction and the presence of noticeable limitations. Lastly, weighted mean values between 1.00 and 1.79 are interpreted as Poor, indicating that the system did not meet user expectations in those areas. This interpretation guide ensures consistent analysis of the survey results and helps determine the system's strengths and areas that require enhancement.

CHAPTER 4

Presentation, Analysis, and Interpretation of Data

This chapter contains presentation, analysis, and interpretation of the data gathered to answer the problems in the study. This part also represent the collective data from forty (40) qualified respondents.

Sub-problem no.1 What are the issues existing in the system in the following aspects:

1.1 Functional Completeness;

Table 2

Problems in the existing system in terms of Functionality

Indicators	Weighted Mean	Interpretation	Rank
1. **Functional Completeness**	4.50	Excellent	1
2. **Functional Correctness**	4.48	Very Good	2.5
3. **Functional Appropriateness**	4.48	Very Good	2.5
Average Computed Mean	4.48	Very Good	

Table 2 assesses the functional suitability of the system whether it fulfills its intended purposes through functional completeness, functional correctness, and functional appropriateness. Scores range from 4.48 to 4.50, with an average computed mean of 4.48, interpreted as "Very Good".

The Functional Suitability evaluated by the two groups of respondents are: The portal covers all the specified tasks and user objectives (wm = 4.50); The provides the correct results with the needed degree of precision (wm = 4.48); The portal functions facilitate the accomplishment of specified tasks and objectives (wm = 4.48).

Hayagan (2022) emphasized that online enrollment has always offered convenience and efficiency, and today's parents are busier than ever. They expect the convenience and ease of submitting their paperwork online, but the benefits of online enrollment systems are not limited to just parents. Administrators and students also benefit from these systems.

1.2 Performance Efficiency;

Table 3

Problems in the existing system in terms of Performance

Indicators	Weighted Mean	Interpretation	Rank
1. **Time Behavior**	4.48	Very Good	1
2. **Resource Utilization**	4.35	Very Good	2.5
3. **Capacity**	4.35	Very Good	2.5
Average Computed Mean	4.39	Very Good	

Table 3 indicates the performance efficiency of the system that evaluate its capability on handling different tasks, composing of three indicators: time behavior, resource utilization, and capacity. Time behavior scored weighted mean of 4.48 while 4.35 for resource utilization and capacity, with an overall average of 4.39 resulting of having a "Very Good" interpretation for the performance efficiency.

Grepon (2021) said that retrieval and generation of data and information is always a concern, as it takes longer

period to consolidate information that is required in the enrollment process so system aids in the improvement of the school's enrollment process and data and information management by integrating a more centralized data storage system and addressing complex transactions by providing specific system modules to address the needs of the school in terms of data storage and retrieval.

1.3 Compatibility;

Table 4

Problems in the existing system in terms of Compatibility

Indicators	Weighted Mean	Interpretation	Rank
1. **Co-existence**	4.40	Very Good	2
2. **Interoperability**	4.50	Excellent	1
Average Computed Mean	4.45	Very Good	

For table 4, it shows how well the system is compatible with different environments, tools, and systems. It has two compatibility indicators which are the interoperability that scored a weighted mean of 4.50 and 4.40 for the co-

existence, with an overall average of 4.45, interpreted as "Very Good".

Gelisana et al. (2022) mentioned that the compatibility of a system or website can be considered as another factor that increases its usefulness.

1.4 Usability;

Table 5

Problems in the existing system in terms of Usability

Indicators	Weighted Mean	Interpretation	Rank
1. **Appropriateness recognizability**	4.55	Excellent	1
2. **Learnability**	4.43	Very Good	3.5
3. **Operability**	4.50	Excellent	2.5
4. **User error protection**	4.35	Very Good	6
5. User interface aesthetics	4.38	Very Good	5
6. Accessibility	4.43	Very Good	3.5
Average Weighted Mean	4.44	Very Good	

Table 5 assesses the online enrollment system's usability through appropriateness recognizability, learnability, operability, user error protection, user interface aesthetics, and accessibility.

The usability indicators being evaluated are: The users can recognize whether the portal is appropriate for their needs. (wm = 4.55); The portal has attributes that make it easy to operate and control (wm = 4.50); The portal can be used by specified users to achieve specified goals of learning to use with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use (wm = 4.43); The portal can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use (wm = 4.43); The portal user interface enables pleasing and satisfying interaction for the user (wm = 4.38); The portal protects users against making errors (wm = 4.35). The scores range from 4.35 to 4.55, with an overall average weighted mean of 4.44, interpreted as "Very Good".

Berro et al. (2025) underscored the pivotal role of usefulness in shaping satisfaction and intention to use,

highlighting the importance of usability in enhancing both usefulness and satisfaction.

1.5 Reliability;

Table 6

Problems in the existing system in terms of Reliability

Indicators	Weighted Mean	Interpretation	Rank
1. Maturity	4.50	Excellent	2
2. Availability	4.58	Excellent	1
3. Fault Tolerance	4.35	Very Good	4
4. Recoverability	4.43	Very Good	3
Average Weighted Mean	4.46	Very Good	

For table 6, it assesses the reliability of the system through indicators like maturity, availability, fault tolerance, and recoverability. The reliability indicators being evaluated are: The portal is operational and accessible when required for use (wm = 4.58); The portal meets needs for reliability under normal operation (wm =

4.50); In the event of an interruption or a failure, the portal can recover the data directly affected and re-establish the desired state of the system (wm = 4.43); The portal operates as intended despite the presence of hardware or software faults (wm = 4.35). The scores range from 4.35 to 4.58 with an overall weighted mean of 4.46, interpreted as "Very Good".

Mina et al.(2021) said that the system's reliability and performance determine its confidence.

1.6 Security;

Table 7

Problems in the existing system in terms of Security

Indicators	Weighted Mean	Interpretation	Rank
1. **Confidentiality**	4.53	Excellent	1.5
2. **Integrity**	4.53	Excellent	1.5
3. **Non-repudiation**	4.33	Very Good	5
4. **Accountability**	4.40	Very Good	4
5. Authenticity	4.48	Very Good	3
Average Weighted Mean	4.45	Very Good	

For table 7, it assesses the security of the system through indicators like confidentiality, integrity, non-repudiation, accountability, and authenticity. The security indicators being evaluated are: The portal ensures that data are accessible only to those authorized to have access. (wm = 4.53); The portal prevents unauthorized access to, or modification of, computer programs or data. (wm = 4.53); The identity of a subject or resource can be proved to be the one claimed.(wm = 4.48); The actions of an entity can be traced

uniquely to the entity. (wm = 4.40); Actions or events can be proven to have taken place so that the events or actions cannot be repudiated later(wm = 4.33). The scores range from 4.33 to 4.53 with an overall weighted mean of 4.45, interpreted as "Very Good".

Villaruz and Diamante (2024) emphasized that a secured system improve user experience and build trust applicants who will use the enrollment system.

1.7 Maintainability;

Table 8

Problems in the existing system in terms of Maintainability

Indicators	Weighted Mean	Interpretation	Rank
1. **Modularity**	4.50	Excellent	2
2. **Reusability**	4.53	Excellent	1
3. **Analysability**	4.40	Very Good	4
4. **Modifiability**	4.43	Very Good	3
5. Testability	4.38	Very Good	5
Average Weighted Mean	4.45	Very Good	

For table 8, it illustrates the system's maintainability and reusability in terms of its modularity, analyzability, modifiability, and testability. The maintainability indicators being evaluated are: an asset can be used in more than one system, or in building other assets. (wm = 4.53); The portal is composed of discrete components such that a change to one component has minimal impact on other components. (wm = 4.50); The portal can be effectively and efficiently modified without introducing defects or degrading existing product quality.(wm = 4.43); Degree of effectiveness and efficiency with which it is possible to assess the impact on the portal of an intended change to one or more of its parts, or to diagnose a product for deficiencies or causes of failures, or to identify parts to be modified. (wm = 4.40); Degree of effectiveness and efficiency with which test criteria can be established for the portal and tests can be performed to determine whether those criteria have been met.(wm = 4.38). The scores range from 4.38 to 4.53 with an overall weighted mean of 4.45, interpreted as "Very Good".

Mina (2021) also stated that a highly maintainable system is one that can be easily modified to adapt to different

situations.

1.8 Portability;

Table 9

Problems in the existing system in terms of Portability

Indicators	Weighted Mean	Interpretation	Rank
1. **Adaptability**	4.48	Very Good	2
2. **Installability**	4.45	Very Good	3
3. **Replaceability**	4.53	Excellent	1
Average Computed Mean	4.48	Very Good	

Table 9 evaluates the system's adaptability, installability, and replaceability. The portability indicators being evaluated are: The portal can replace another specified software product for the same purpose in the same environment (wm = 4.53); The portal can replace another specified software product for the same purpose in the same environment (wm = 4.48); Degree of effectiveness and efficiency with which the portal can be successfully

installed and/or uninstalled in a specified environment (wm = 4.45). The score range from 4.45 to 4.53, with an overall weighted mean 4.48, interpreted as "Very Good".

Balcita and Palaoag (2020) introduced a school management system that is a highly portable which is easily accessible on multiple devices and platforms which aid the conflicts present in managing the manual records of the school.

Sub-problem no.2 Based on the data gathered, what application/website may be developed?

The online enrollment system was developed based on the data gathered from IT/CS students and IT Professionals. Key features include online registration for enrollment, online payment of tuition fee, checking of enrollment status, and management of enrollees on the administrative side of the system.

FLOWCHART

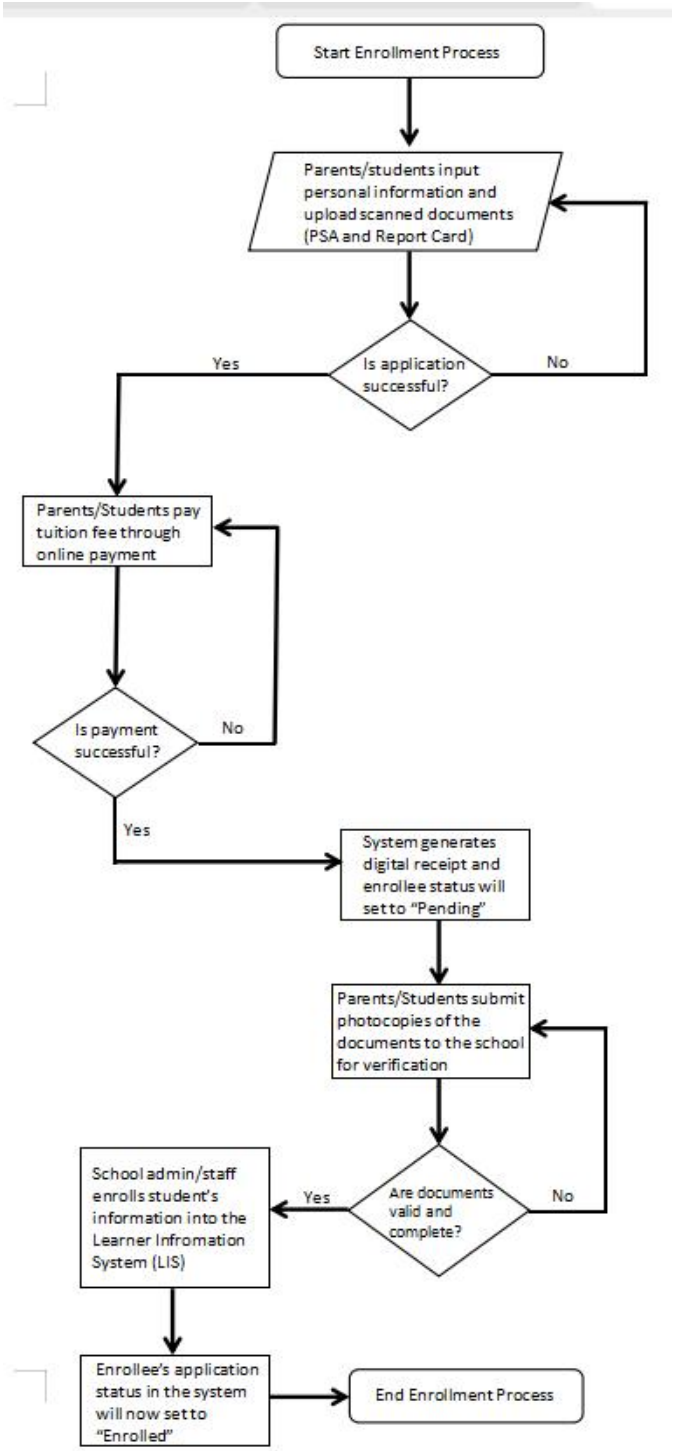

Figure 2 Flowchart of the Online Enrollment System

This figure shows the flowchart of the online enrollments system, which visually represents the step-by-step process of the whole system. It illustrates the process, starting from the online registration through the document verification and ending with approved enrollment registration and storage. The flowchart provides a clear depiction of how users will use the whole system and how the system will serve the students, parents, and school staffs.

Data Flow Diagram

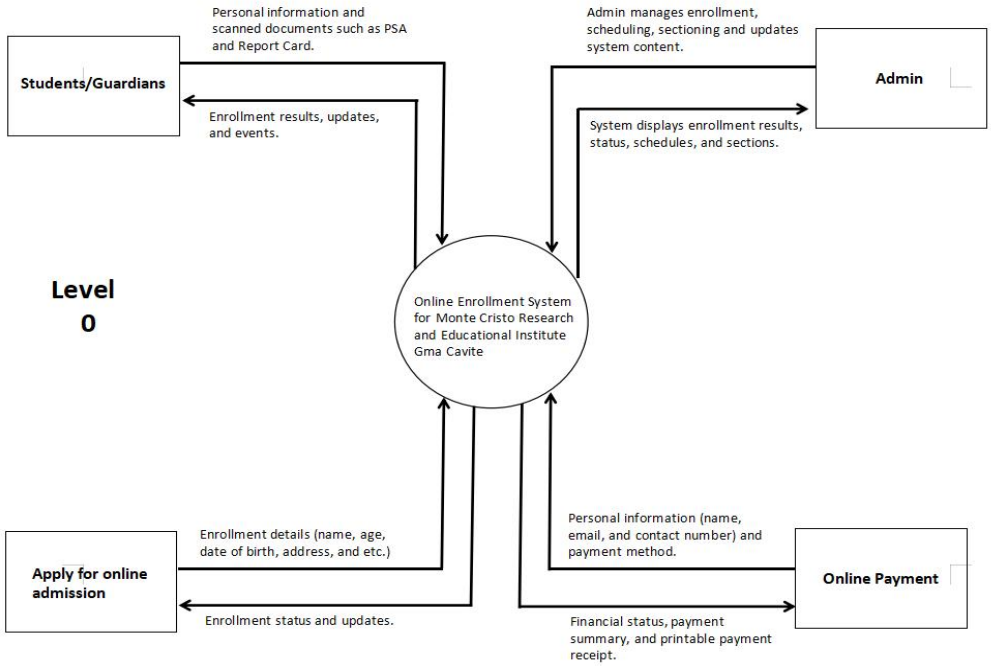

Figure 3 Conceptual Diagram Level 0 of the Online Enrollment System

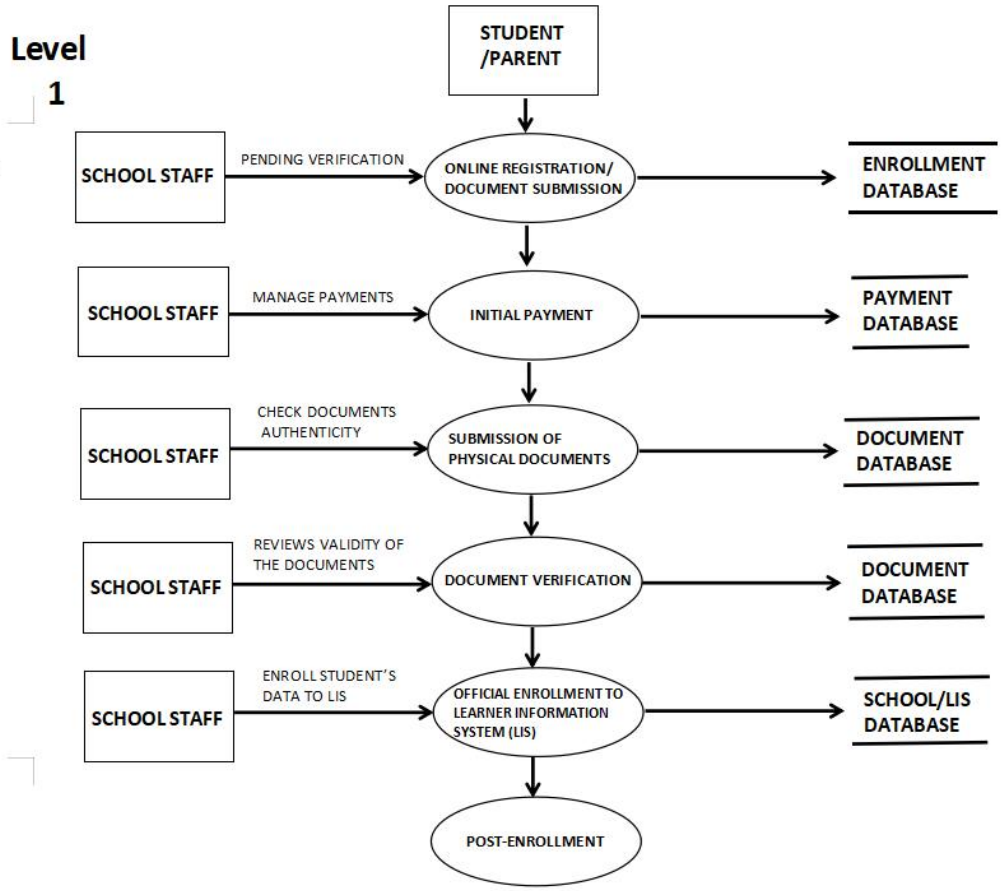

Figure 4 Conceptual Diagram Level 1 of the Online Enrollment System

This figure illustrates the highest level of interaction between users and the online enrollment system. It illustrates the process flow of how students/parents will use the system for online enrollment.

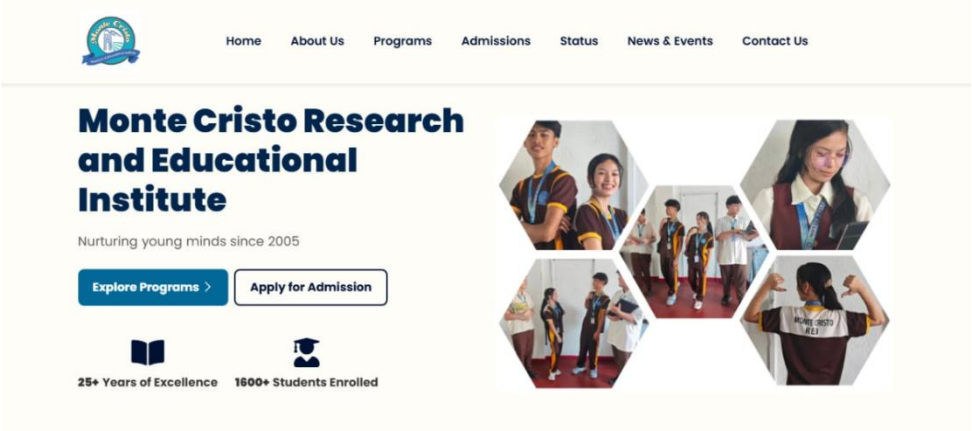

Figure 5. HOMEPAGE

Figure 5 shows the first part of the system that the user will see. It includes the about, programs, admissions, status, news and events, and contact page buttons so the user can navigate to that page easily.

Enroll Your Child

Join our family of learners for the upcoming academic year

Please fill out the form below to start the enrollment process. Print legibly all information required in CAPITAL letters.

Requirements: *You will typically need to submit the following required documents based on the grade level:*

- ***For Kindergarten:*** *Photocopy of NSO/PSA Birth Certificate.*
- ***For Incoming Grade 1:*** *Photocopy of NSO/PSA Birth Certificate AND Kindergarten Report Card.*
- ***For Grade 2-6:*** *Previous Report Card.*
- *Other supporting documents may be requested.*

All applications remain in pending status until the original documents are submitted to the school within the specified enrollment period.

Please ensure all uploaded files are clear and readable.

Basic Enrollment Information

School Year * 2025-2026

Grade Level Applying For * Select Grade Level

Returning (Balik-Aral) * No

Learner Information

With LRN? * ○ Yes ◉ No

Last Name * **First Name *** **Middle Name**

Extension Name (Jr., III, etc.) **Date of Birth *** mm/dd/yyyy **Age * (Auto-calculates)** **Sex *** Select Sex

Place of Birth (Municipality/City) * **Mother Tongue ***

Belonging to any Indigenous Peoples (IP) Community? * ○ Yes ◉ No

If Yes, please specify:

Is your family a beneficiary of 4Ps? * ○ Yes ◉ No

If Yes, write the 4Ps Household ID Number:

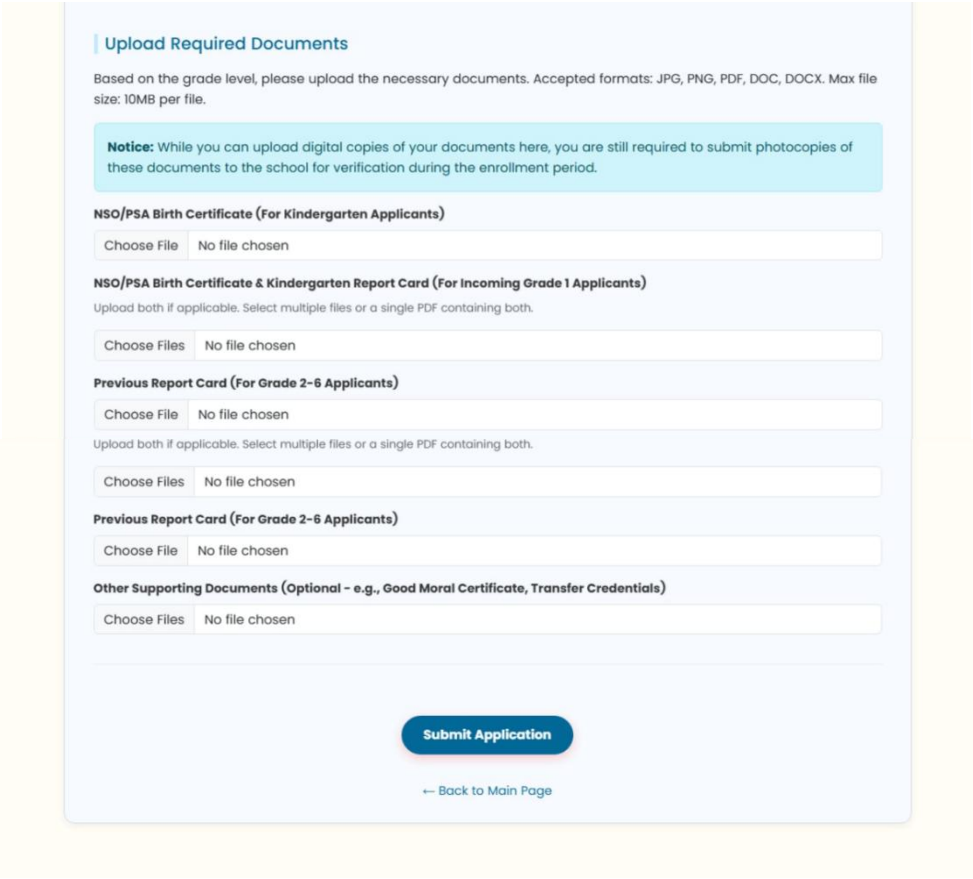

Figure 6. ONLINE ENROLLMENT

Figure 6 shows the systems online enrollment feature. It includes all the information and documents that are needed in the enrollment process.

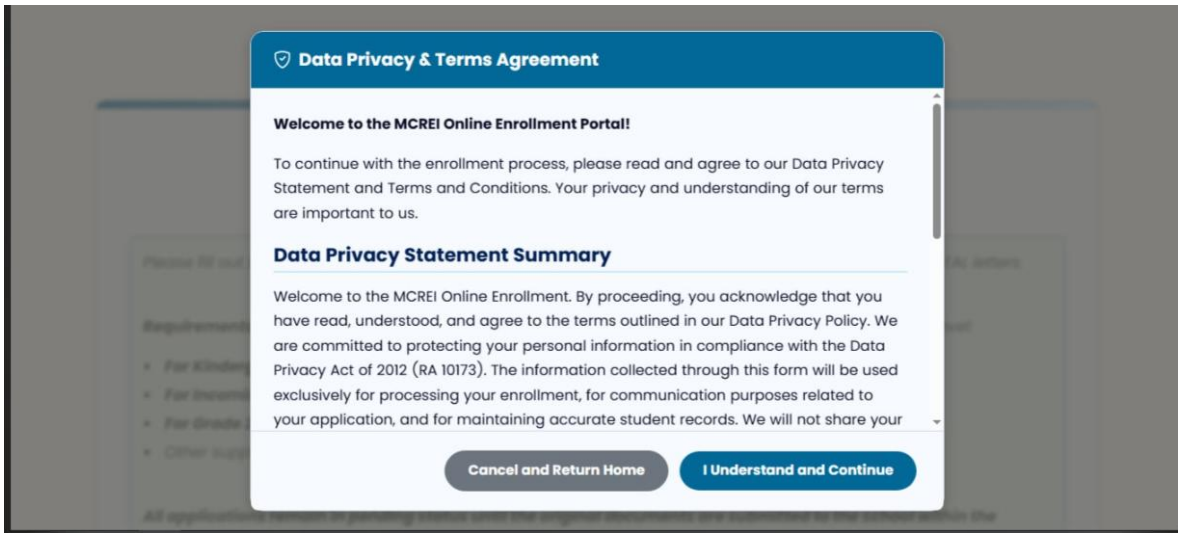

Figure 7. DATA PRIVACY AND TERMS AGREEMENT

Figure 7 shows the Data Privacy and Terms Agreement before that will pop-up before navigating to the online enrollment form ensuring that the user will understand and gree to the terms outlined in the Data Privacy Policy of the system.

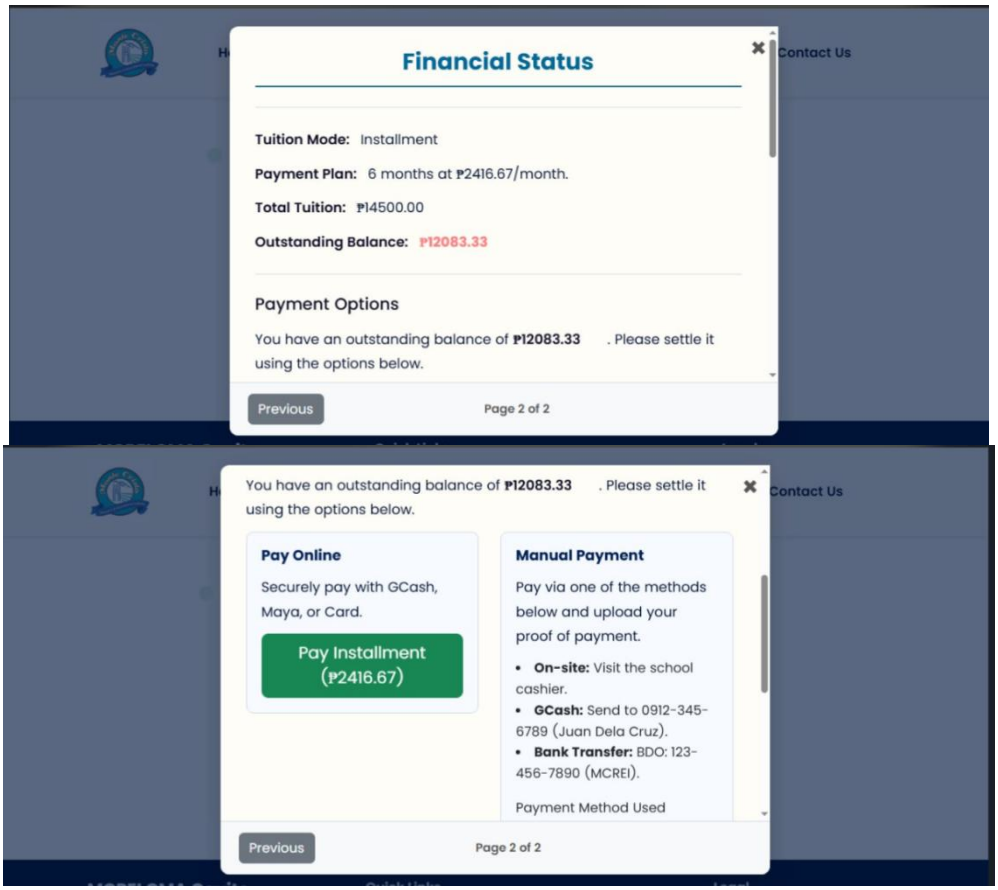

Figure 8. FINANCIAL STATUS

Figure 8 shows the financial status of the student with the options of paying their outstanding balance through online or a manual payment.

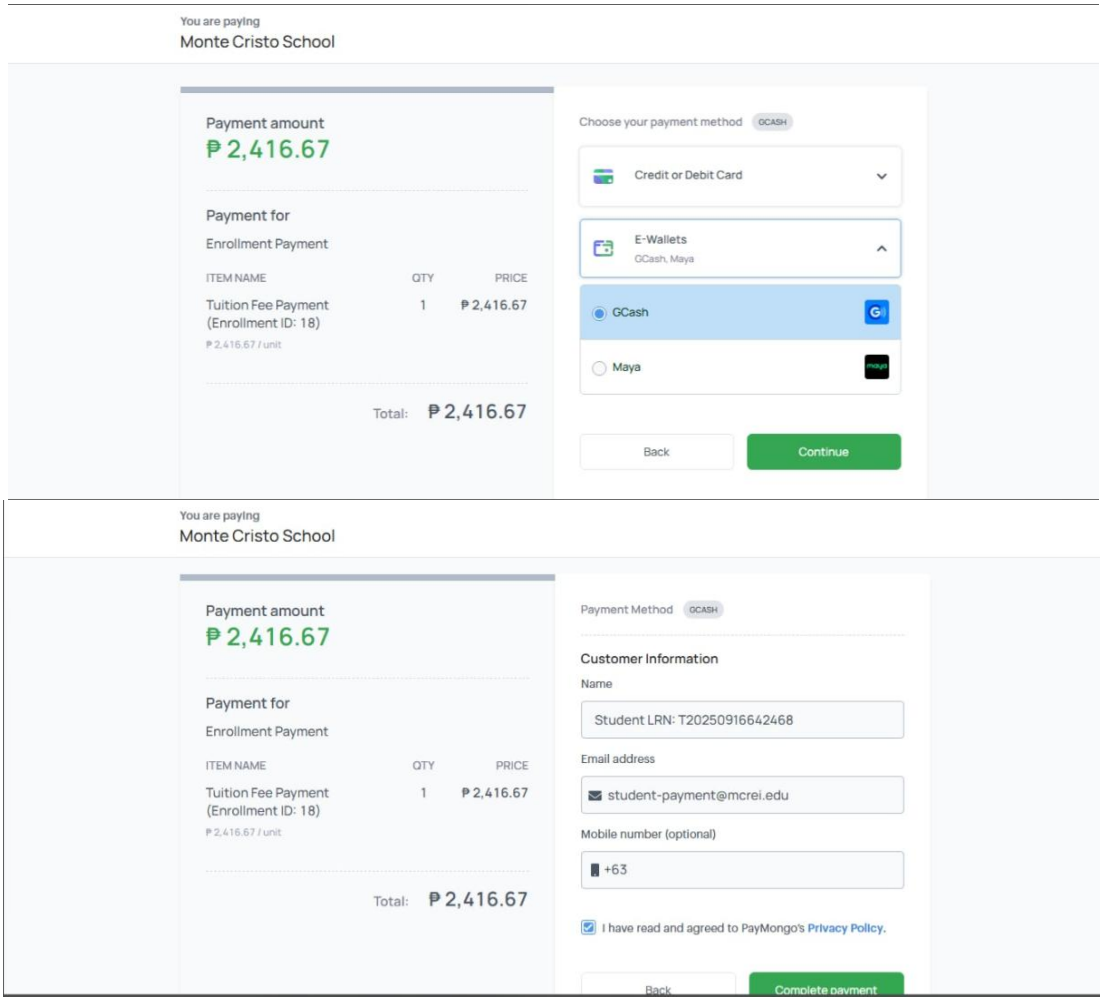

Figure 9. PAYMENT GATEWAY

Figure 9 shows on how the user will settle their payment. It consists of the payment method option and the student/parent information.

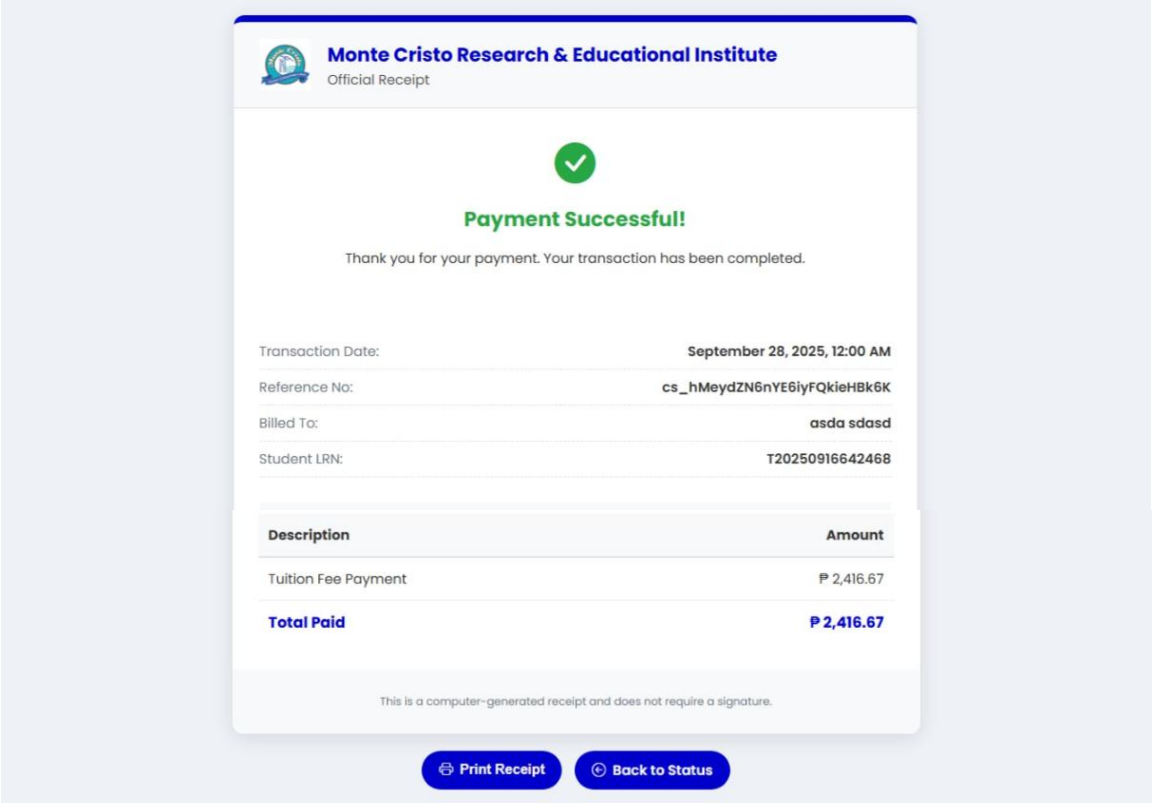

Figure 10. DIGITAL RECEIPT

Figure 10 shows the generation of digital receipt after a successful payment.

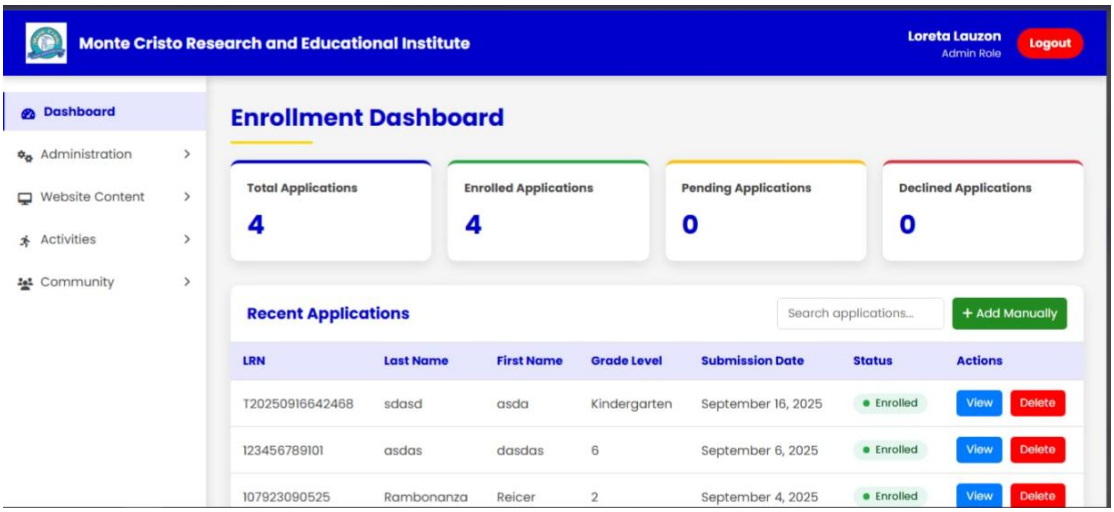

Figure 11. ADMIN PAGE

Figure 11 shows the admin page where the school admin can manage the entire enrollment process and as well as modify the system's content.

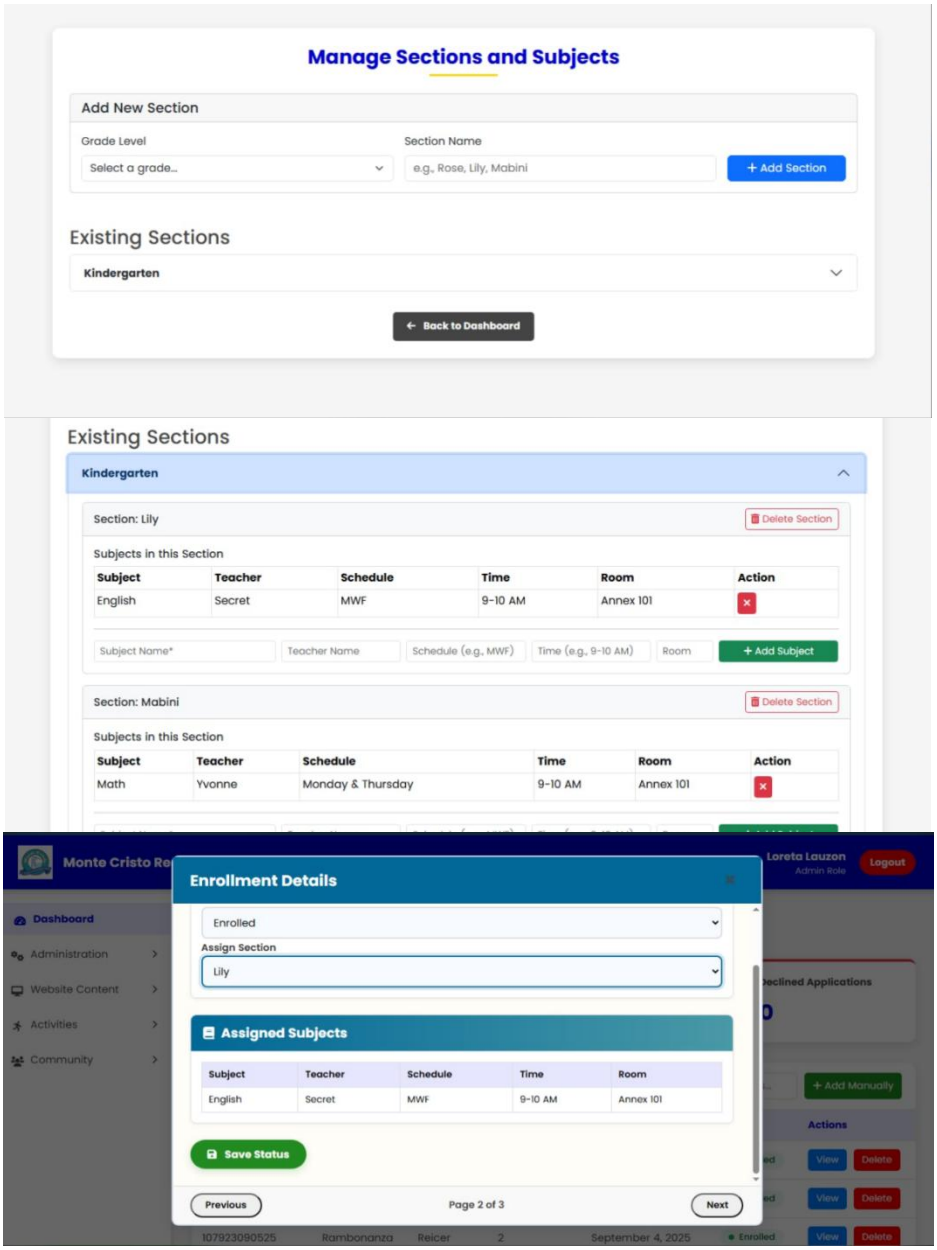

Figure 12. SECTION AND SUBJECT MANAGEMENT

Figure 12 shows the section where the admin can manage the subject and schedule of the students and teachers.

Sub-problem no.3 How do evaluators assess the developed website as to:

3.1 Functional Completeness;

Table 10

Assessment of the developed system in terms of Functional Completeness

Functional Completeness	IT Professional Mean	VI	IT Student Mean	VI	CM	VI	Rank
1. The portal covers all the specified tasks and user objectives.	4.41	Very Good	4.46	Very Good	4.50	Excellent	1
2. Functional Correctness :The provides the correct results with the needed degree of precision.	4.41	Very Good	4.46	Very Good	4.48	Very Good	2.5
3. Functional Appropriateness: The portal functions facilitate the accomplishment of specified tasks and objectives	4.42	Very Good	4.46	Very Good	4.48	Very Good	2.5
Average Computed Mean	**4.41**	**Very Good**	**4.46**	**Very Good**	**4.48**	**Very Good**	

Based on Table 10, the evaluator found out that the system can perform its function on its intended purpose, having a score of 4.50, interpreted as "Excellent" while functional correctness and functional appropriateness scored 4.48, implying that the system is really highly functional.

Grepon et al. (2021) introduced a school management system that provides complete functionality that is accurate and appropriate addressing the issues facing in manual process of managing school records.

3.2 Performance Efficiency;

Table 11

Assessment of the developed system in terms of Performance

Performance Efficiency	IT Profes-sional Mean	VI	IT Student Mean	VI	CM	VI	Rank
1. Time Behaviour: response and processing times and throughput rates of the portal, when performing its functions, meet requirements.	4.41	Very Good	4.46	VeryGood	4.48	VeryGood	1
2. Resource Utilization: The amounts and types of resources used by the portal, when performing its functions, meet requirements.	4.40	VeryGood	4.46	VeryGood	4.35	VeryGood	2.5
3. Capacity: The maximum limits of the portal parameter meet requirements	4.40	VeryGood	4.46	VeryGood	4.35	VeryGood	2.5

Average Computed Mean	**4.41**	**VeryGood**	**4.46**	**VeryGood**	**4.39**	**VeryGood**	

Based on Table 11, the evaluator found out that the system has an excellent performance efficiency, having an overall computed mean of 4.39, interpreted as "Very Good." This high score implies that the system is well-optimized, demonstrating optimal response and processing times (Time Behavior) and appropriate resource utilization. Furthermore, the consistent rating affirmed the system's capacity to handle its maximum workload and maintain stable performance throughout different usage conditions.

Abisado et al. (2020) said that due to large population of enrollee reaching 3000 applicants, manual setup for data management is not an easy task that is why he addressed the importance of having an online enrollment system addressing those kinds of issues in schools.

3.3 Compatibility;

Table 12

Assessment of the developed system in terms of Compatibility

Compatibility	IT Professional Mean	VI	IT Student Mean	VI	CM	VI	Rank
1.Co-existence: The portal can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product.	4.41	VeryGood	4.47	VeryGood	4.40	VeryGood	2
2.Interoperability : The portal can exchange information and use the information that has been exchanged.	4.42	VeryGood	4.47	VeryGood	4.50	Excellent	1
Average Computed Mean	**4.41**	**VeryGood**	**4.47**	**VeryGood**	**4.45**	**VeryGood**	

Based on Table 12, the system's compatibility received a strong ratings, with the overall mean interpreted as "Very Good." This high score confirms that the system is highly capable of integration and interoperability. The fact that IT professionals and students rated the system almost identically further validates that it can seamlessly share resources and exchange information when working with other systems, ensuring a smooth and effective operation across different technology platforms.

Villaruz and Diamante (2024) also pointed out that hardware independence of a system or compatibility across platforms is an indicator of a strong and reliable system.

3.4 Usability;

Table 13

Assessment of the developed system in terms of Usability

Usability	IT Profes-sional Mean	VI	IT Student Mean	VI	CM	VI	Rank
1. Appropriateness recognizabilty: The users can recognize whether the portal is appropriate for their needs.	4.41	VeryGood	4.47	VeryGood	4.55	Excellent	1
2. Learnability: The portal can be used by specified users to achieve specified goals of learning to use with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.	4.41	VeryGood	4.46	VeryGood	4.43	VeryGood	3.5

Table 13, continued

Usability	IT Profes-sional Mean	VI	IT Student Mean	VI	CM	VI	Rank
3. Operability: The portal has attributes that make it easy to operate and control.	4.40	VeryGood	4.46	VeryGood	4.50	Excellent	2
4. User error protection: The portal protects users against making errors	4.39	VeryGood	4.47	VeryGood	4.35	VeryGood	6
5. User interface aesthetics: The portal user interface enables pleasing and satisfying interaction for the user.	4.39	VeryGood	4.47	VeryGood	4.38	VeryGood	5
6. Accessibility: The portal can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.	4.41	VeryGood	4.47	VeryGood	4.43	VeryGood	3.5

Average Computed Mean	**4.40**	VeryGood	**4.47**	VeryGood	**4.44**	VeryGood	

Based on Table 13, the evaluator found out that the system was highly usable, having an overall computed mean of 4.44, interpreted as "Very Good." This high score indicates the system possesses a strong user-centered design, with the highest rating in Appropriateness Recognizability (cm = 4.55) confirming the portal meets user expectations. Furthermore, the ratings for Operability and Accessibility demonstrate the system is intuitive, manageable, and inclusive for a wide user base, ensuring a positive and efficient user experience.

Bagacay et al. (2024) discussed that a usable system that has user-friendly interface, intuitive design, and time-saving capabilities can highly satisfy various users.

3.5 Reliability;

Table 14

Assessment of the developed system in terms of Reliability

Reliability	IT Profes-sional Mean	VI	IT Student Mean	VI	CM	VI	Rank
1. Maturity: The portal meets needs for reliability under normal operation.	4.40	VeryGood	4.48	VeryGood	4.50	Excellent	2
2. Availability: The portal is operational and accessible when required for use	4.40	VeryGood	4.47	VeryGood	4.58	Excellent	1
3. Fault tolerance: The portal operates as intended despite the presence of hardware or software faults	4.39	VeryGood	4.47	VeryGood	4.35	VeryGood	4
4. Recoverability: in the event of an interruption or a failure, the portal can recover the data directly affected and re-establish	4.39	VeryGood	4.47	VeryGood	4.43	VeryGood	3

the desired state of the system.							
Average Computed Mean	**4.39**	VeryGood	**4.47**	**VG**	**4.46**	VeryGood	

The data from Table 14 confirms the system's high reliability, achieving an overall mean of 4.46, interpreted as "Very Good." Significantly, the system excels in Availability (cm = 4.58, Excellent), demonstrating that the portal is consistently operational and ready for user access whenever needed. The strong score in Maturity (cm = 4.50, Excellent) further assures that the system performs dependably under normal operating conditions. Additionally, solid ratings in both Recoverability and Fault Tolerance (Very Good interpretations) prove the system's resilience and it can effectively restore functions after interruptions and maintain stability even when faced with technical issues.

Mag-aso et al. (2025) proposed an online enrollment system with a high rating in terms of its reliability, implying that an online enrollment system is a beneficial and reliable to students, parents, and school staffs.

3.6 Security;

Table 15

Assessment of the developed system in terms of Security

Security	IT Profes-sional Mean	VI	IT Student Mean	VI	CM	VI	Rank
1.Confidentiality: The portal ensures that data are accessible only to those authorized to have access.	4.39	VeryGood	4.48	VeryGood	4.53	VeryGood	1.5
2.Integrity: The portal prevents unauthorized access to, or modification of, computer programs or data.	4.38	VeryGood	4.47	VeryGood	4.53	Excellent	1.5
3. Non-repudiation: actions or events can be proven to have taken place so that the events or actions cannot be repudiated later.	4.36	VeryGood	4.47	VeryGood	4.33	VeryGood	5
4. Accountability: the actions of an entity can be traced uniquely to the entity.	4.38	VeryGood	4.48	VeryGood	4.40	VeryGood	4
5. Authenticity: the identity of a subject or resource can be proved to be the one claimed.	4.40	VeryGood	4.48	VeryGood	4.48	VeryGood	3

Average Computed Mean	**4.38**	VeryGood	**4.48**	VeryGood	**4.45**	VeryGood	

The security assessment in Table 15 yielded strong results, with the overall computed mean of 4.45 interpreted as "Very Good." This high rating confirms the system's commitment to protecting user data. The indicators for Confidentiality and Integrity were rated highest (cm = 4.53, Excellent), demonstrating that the portal effectively restricts unauthorized access and prevents data tampering. Furthermore, the system is adept at verifying user identity (Authenticity, Very Good) and ensuring that all actions are tracked and verifiable (Accountability and Non-repudiation, both Very Good). The agreement between IT professionals and students on these metrics underscores that the system upholds robust security measures, establishing a reliable and protected environment for all user information.

Villaruz and Diamante (2024) stated how important to adopt strong security protocols such as secure data storage practices and encryption techniques, safeguarding applicant information.

3.7 Maitainability;

Table 16

Assessment of the developed system in terms of Maintainability

Maintainability	**IT Profes-sional**	**VI**	**IT Student**	**VI**	**CM**	**VI**	**Rank**
	Mean		**Mean**				
1. Modularity: The portal is composed of discrete components such that a change to one component has minimal impact on other components	4.40	VeryGood	4.48	VeryGood	4.50	Excellent	2
2. Reusability: an asset can be used in more than one system, or in building other assets.	4.40	VeryGood	4.47	VeryGood	4.53	Excellent	1

Table 16, continued

Maintainability	**IT Profes-sional**	**VI**	**IT Student**	**VI**	**CM**	**VI**	**Rank**
	Mean		**Mean**				
3. Analysability: Degree of effectiveness and efficiency with which it is possible to assess the impact on the portal of an intended change to one or more of its parts, or to diagnose a product for deficiencies or causes of failures, or to identify parts to be modified	4.38	VeryGood	4.46	VeryGood	4.40	VeryGood	4
4. Modifiability: the portal can be effectively and efficiently modified without introducing defects or degrading existing product quality	4.38	VeryGood	4.47	VeryGood	4.43	VeryGood	3

Table 16, continued

Maintainability	IT Profes-sional Mean	VI	IT Student Mean	VI	CM	VI	Rank
5. Testability: Degree of effectiveness and efficiency with which test criteria can be established for the portal and tests can be performed to determine whether those criteria have been met.	4.43	VeryGood	4.47	VeryGood	4.38	VeryGood	5
Average Computed Mean	**4.40**	VeryGood	**4.47**	VeryGood	**4.45**	VeryGood	

The evaluation in Table 16 confirms the system's high marks for maintainability, achieving an overall mean of 4.45, which is interpreted as "Very Good." This excellent outcome reflects a system that is fundamentally well-structured and easy to manage over time. The highest scores were attained in Reusability (cm = 4.53) and Modularity (cm = 4.50), both rated "Excellent," which essentially means the system is built efficiently and its parts can be reused in future

projects, and individual components can be updated independently without breaking the whole. Consistent "Very Good" ratings across Analysability, Modifiability, and Testability further demonstrate that the system can be thoroughly understood, modified, and validated with ease, ensuring smooth, stable improvements for its long-term viability.

Ronquillo et al. (2022) presented an online management system that is a modifiable in situations where new functionalities may be required or requested by the users, highlighting the flexibility as an attribute and that it can make data stored accessible by more than one user at different times for different purposes.

3.8 Portability;

Table 17

Assessment of the developed system in terms of Portability

Portability	IT Profes-sional Mean	VI	IT Student Mean	VI	CM	VI	Rank
1. Adaptability: The portal can replace another specified software product for the same purpose in the same environment	4.47	VeryGood	4.49	VeryGood	4.48	VeryGood	2
2. Installability: Degree of effectiveness and efficiency with which the portal can be successfully installed and/or uninstalled in a specified environment	4.45	VeryGood	4.50	Excellent	4.45	VeryGood	3
3. Replaceability: The portal can replace another specified software product for the same purpose in the same environment	4.50	Excellent	4.53	Excellent	4.53	Excellent	1
Average Computed Mean	**4.47**	VeryGood	**4.51**	**Excellent**	**4.48**	VeryGood	

Based on Table 17, the system's portability was assessed as "Very Good," with an overall computed mean of 4.48. This high score confirms that the portal performs reliably across diverse environments and conditions. A key strength is Replaceability (cm = 4.53), which achieved an "Excellent" rating, suggesting the system can seamlessly take the place of similar platforms without performance loss. The consistently high ratings for both Adaptability and Installability (Very Good interpretations) further demonstrate the system's flexibility and it adjusts efficiently to new platforms and can be installed or uninstalled smoothly. Overall, these findings highlight the portal's strong portability, assuring users of its flexibility and broad compatibility in various usage scenarios.

Tagasa et al. (2022) pointed out that highly portable system allows diverse range of end-users.

Sub-problem no.4 For the findings, what implementation program may be proposed?

PHASE 1: IMPLEMENTATION STAGE

Activity	Description	Responsible Person	Time	Output	Budget
System Installation	Install the system on the main server and set the settings for use.	IT Staff	Week 1	System working	₱10,000 (includes hosting)
Data Migration	Move student and enrollment records to the new system.	IT Staff	Week 2	Data moved and checked	₱2,500
User Account Setup	Make admin accounts	IT Staff	Week 2	Accounts ready	None (manual setup)
Interface Configuration	Change the front part and design to match the school style.	UI/UX Designer	Week 3	New interface	₱5,000
Deployment Review	Check all parts work before launch.	QA Team, IT Staff	Week 4	Report of readiness	₱5,000

PHASE 2: TESTING AND INTEGRATION

Activity	Description	Responsible Person	Time	Output	Budget
System Testing	Test how the system works and fix problems.	QA Team, Development Team	Week 5	Test report with fixes	₱5,000
User Acceptance Testing	Let main users try the system to see if it is easy to use.	Technology staff and chosen users	Week 6	Approval document	None
Integration Testing	Make sure parts work together and connect.	IT Staff	Week 6	Working system links	₱3,000
Security Evaluation	Check for safety and protect data.	Security specialist, IT Staff	Week 7	Security report	₱10,000
Documentation Review	Update guides based on test results.	Project team, IT Staff	Week 7	Updated guides	₱8,000

PHASE 3: MAINTENANCE AND SUPPORT

Activity	Description	Responsible Person	Time	Output	Budget
Performance Monitoring	Watch system activity to keep it efficient.	IT staff	Monthly	Activity reports	₱1,000 per report
Routine Updates	Put software and safety updates when needed.	IT Staff	Quarterly	Updated system	₱5,000 per quarter
User Support	Help users and solve problems.	IT Staff	Ongoing	Solved issues	₱1000 per issue
Backup Management	Make copies of data and test restores.	IT Staff	Weekly	Backup logs	₱3,000 yearly
System Enhancement	Add improvements and new features.	Development team	As needed	Improved system	₱4,000 per update

The implementation plan for the Online Enrollment System ensures that it is properly introduced, tested, and maintained to support a smooth and efficient enrollment

process. It details the necessary steps to prepare the system for active use, verify that all its features function correctly and securely, and maintain its performance through continuous monitoring, updates, and technical support. This plan aims to ensure the system's reliability, ease of use, and long-term effectiveness in improving the enrollment procedures and communication between students, school staff, and administrators.

Chapter 5

Summary, Conclusions, and Recommendations

Summary

The findings of this study are summarized as follows.

1. What are the problems/issues in the existing system as to;

The assessment confirmed the proposed system's exceptional core function, earning an overall "Very Good" rating for its functional completeness. Evaluators found the system's ability to perform its intended function to be "Excellent," giving it a weighted mean of 4.50. The system's functional correctness and appropriateness also scored very high at 4.48. Essentially, this shows the system doesn't just work but works exactly as it should and is the right solution for its purpose.

For performance efficiency, the system was rated highly, achieving an overall computed mean of 4.39, interpreted as "Very Good." This high score indicates the system is well-optimized, demonstrating consistency across all measured indicators. Time Behaviour, which tracks response and

processing times, ranked highest with a computed mean (CM) of 4.48. Also, strong ratings were observed for both Resource Utilization (CM = 4.35) and Capacity (CM = 4.35), confirming that the system appropriately uses available resources and can effectively handle its maximum workload while maintaining stable performance.

For compatibility, the system achieved strong results, with the overall computed mean of 4.45 interpreted as "Very Good." This high rating confirms the system's ability to seamlessly integrate and function within different technological environments. The highest-ranked indicator was Interoperability, achieving an "Excellent" computed mean of 4.50, which demonstrates the portal's effectiveness in exchanging and utilizing shared information. Furthermore, Co-existence followed closely with a score of 4.40$, indicating the portal performs its functions efficiently while sharing common resources without negatively impacting other systems.

For usability, the system achieved a strong result, with the overall computed mean of 4.44 interpreted as "Very Good."

This high score confirms the system has a strong user-centered design and is easy to use. The highest-ranked indicator was Appropriateness Recognizability, scoring an "Excellent" computed mean of 4.55, which ensures users readily perceive the system as meeting their needs. Operability also received an "Excellent" rating (4.50), showing the system is easy to control. Furthermore, indicators for Learnability and Accessibility were rated highly, demonstrating the system's intuitive design and inclusiveness for a wide range of users, ensuring an efficient and positive user experience.

For reliability, the system's assessment result yielded a strong overall computed mean of 4.46, interpreted as "Very Good." This high rating confirms the system's dependable and resilient performance. The highest-ranked indicator was Availability, scoring an "Excellent" computed mean of 4.58, ensuring the portal is consistently operational and accessible when needed. Maturity also achieved an "Excellent" rating (4.50), confirming reliable performance under normal operations. Also, solid ratings were observed for both

Recoverability and Fault Tolerance (scored 4.43 and 4.35, respectively), which collectively demonstrate the system's ability to restore data after interruptions and maintain stability even when faced with technical faults.

For security, the system's assessment result in a strong overall computed mean of 4.45, interpreted as "Very Good." This high rating confirms the system upholds robust measures for protecting user information. The highest-ranked indicators were Confidentiality and Integrity, both scoring an "Excellent" computed mean of 4.53. This demonstrates that the system is highly effective at restricting access to authorized users and preventing unauthorized data modification. Furthermore, strong "Very Good" ratings for Authenticity (4.48), Accountability (4.40), and Non-repudiation (4.33) collectively ensure that user identities are verified and that all actions are uniquely traceable and verifiable, establishing a reliable and protected environment for users.

For maintainability, the system's Maintainability was

assessed highly, resulting in an overall computed mean of 4.45, interpreted as "Very Good." This result confirms the system is well-structured and easy to adapt over time. The highest-ranked indicator was Reusability, scoring an "Excellent" computed mean of 4.53, which means system components can be efficiently leveraged in future developments. Modularity also achieved an "Excellent" rating (4.50), assuring that changes to one part of the system will have minimal impact on others. Moreover, solid "Very Good" ratings across Analysability, Modifiability, and Testability demonstrate that the system can be easily understood, updated, and validated, ensuring stable and efficient long-term improvement.

Lastly, for portability, the system was assessed as highly capable, achieving an overall computed mean of 4.48, which is interpreted as "Very Good." It implies that the system can perform reliably across diverse environments and conditions. The highest-ranked indicator was Replaceability, scoring an "Excellent" computed mean of 4.53. This suggests the portal can seamlessly take the place of other similar systems without issues or performance loss. "Very Good"

ratings were also observed for Adaptability (4.48) and Installability (4.45), collectively demonstrating the system's flexibility to adjust to different platforms and its ease of installation and uninstallation, assuring broad compatibility in various usage scenarios.

2. Based on the data gathered, what application/website may be developed?

Based on the data gathered, the online enrollment system was developed to address identified issues in the school's enrollment process. The system features include online registration, online payment of tuition fees, online approval of the applicant's information and personal documents, and a user-friendly design to meet the needs of users and administrator. Key functionalities, such as a tracking of enrollment status, generation of Certificate of Registration, news and event, and programs were included to streamline flow of information and enrollment process and improve efficiency. The features were incorporated to make sure that the user will use the system conveniently and efficiently.

3. How do evaluators assess the developed application / website as to:

The comprehensive quality assessment confirms the Online Enrollment System is a high-performing product, consistently achieving an overall interpretation of "Very Good" across all eight measured quality characteristics. The system's Functional Suitability to its intended purpose is exceptionally strong, reflected by its Excellent rating for performing the intended function (4.50). This finding establishes that the system not only works reliably but is precisely the right tool for the job, validated by similarly high scores for functional correctness and appropriateness (both 4.48).

Performance Efficiency achieved a "Very Good" overall computed mean of 4.39. This high score indicates the system is well-optimized for speed and resource management, with Time Behaviour, tracking response and processing times, securing the highest individual computed mean of 4.48, followed by resource utilization and capacity, with a computed mean of 4.35.

With a "Very Good" computed mean of 4.45, Compatibility demonstrated superior capability for seamless integration. The evaluation highlighted Interoperability as a key strength, earning an Excellent rating (4.50), confirming the system effectively exchanges information with other tools and platforms, followed by Co-existence (4.40)

Usability was affirmed as being highly effective and user-centered, reflected by its "Very Good" mean of 4.44. This quality is best exemplified by the Excellent result for Appropriateness Recognizability (4.55), ensuring users immediately understand that the portal meets their needs, followed by Operability (4.50), Learnability (4.43), Accessibility (4.43), User Interface Aesthetics (4.38), User Error Protection (4.35).

The system shows clear dependability and resilience, earning a "Very Good" mean of 4.46 for Reliability. Its consistently high performance is anchored by Availability, which secured the highest score across all characteristics at

an Excellent 4.58, ensuring the portal is operational whenever required, followed by Maturity (4.50), Recoverability (4.43), and Fault Tolerance (4.35) which is the lowest among all the indicators.

Protection of user data is paramount, reflected by the "Very Good" mean of 4.45 for Security. Both Confidentiality and Integrity were rated Excellent (4.53), which demonstrates the portal's effectiveness in restricting unauthorized access and preventing data tampering, followed by Authenticity (4.48), Accountability (4.40), and Non-repudiation (4.33).

The portal is well-structured for long-term support, achieving a "Very Good" mean of 4.45 in Maintainability. This strength is largely due to Reusability (4.53), which received an Excellent rating and confirms components can be efficiently leveraged in future developments, followed by Modularity (4.50), Modifiability (4.43), Analysability (4.40), and Testability (4.38).

Finally, the system's flexibility to operate across

different environments is affirmed by its Portability "Very Good" mean of 4.48. This capability is led by an Excellent score in Replaceability (4.53), suggesting the portal can seamlessly substitute similar systems without performance loss, which followed by Adaptability (4.48), and Installability (4.45).

While the system's performance is universally strong, the evaluation indicates minor opportunities for enhancement in the lowest-rated indicators. These include Fault Tolerance (4.35), User Error Protection (4.35), and Non-repudiation (4.33). Addressing these specific areas would further elevate the system's already robust safeguards against technical and human errors.

4. From the findings, what implementation programs may be proposed?

The implementation of the online enrollment system is structured into three key phases to ensure an organized rollout and long-term sustainability. The first phase focuses on initial deployment, which includes installing the system

on the main server, moving existing student records into the new database, setting up user accounts, customizing the interface, and verifying readiness before launch. These activities are scheduled across the first four weeks and are carried out by IT staff, database personnel, developers, and quality assurance members. The second phase centers on testing and integration to guarantee reliability before full use. This stage covers system functionality checks, user trials to assess ease of use, compatibility testing between modules, security evaluations, and updates to documentation. These tasks take place between weeks five and seven and involve technical staff, selected users, security specialists, and the project team. The final phase ensures the system remains effective over time through continuous maintenance and support. Regular monitoring, software updates, user assistance, data backups, and feature improvements are performed on a scheduled or as-needed basis. This ongoing phase supports system performance, user satisfaction, and future developments. Overall, the program provides a clear path from deployment to long-term operation, ensuring the system is properly installed, thoroughly tested, and

continuously improved.

Conclusions

1. While the system's performance is consistently strong, the assessment did pinpoint specific areas where future refinement would be most beneficial. The indicators that scored lowest were Non-repudiation (in Security), Fault Tolerance (in Reliability), and User Error Protection (in Usability). Focusing on these specific points will help strengthen the system's ability to resist technical failures, ensure full accountability for user actions, and offer better protection against mistakes, ultimately making an already robust system even more secure and stable.

2. The online enrollment system was developed as a modern solution to the identified gaps in school's enrollment processes. It includes beneficial and valuable features online registration, online payment for tuition fees, enrollment status tracking, and a user-friendly interface, specifically designed to enhance the overall user experience.

3. The developed Online Enrollment System was thoroughly assessed by evaluators and achieved a strong overall rating of "Very Good" across all eight quality characteristics. This

outcome clearly reflects significant improvements over the existing system's capabilities. Specifically, the system now demonstrates superior Functional Suitability, ensuring it is precisely the right tool for enrollment needs. Its Performance Efficiency has been dramatically enhanced, yielding optimal response and processing times. Compatibility now confirms seamless integration with other tools, while Usability provides an intuitive and accessible user experience. The system's Reliability is assured by its consistent availability, and its Security is robustly improved through strong confidentiality and integrity measures. Furthermore, the system's long-term Maintainability is secured by highly reusable components, and its Portability guarantees flexibility across various operating environments.

4. The researchers concluded that the strategy for implementing the Online Enrollment System was highly effective. This plan ensures an effective approach to launching the system, rigorous testing of all features, and reliable, continuous maintenance, all of which are essential for addressing diverse user needs with a high level of efficiency.

Recommendations

The following recommendations were formulated by the researchers to maintain and improve the Online Enrollment System.

1. The researchers suggests improving the setup and use of system resources to boost the performance of the online enrollment platform. By making the system run more efficiently, it can speed up processing, lessen technical issues, and lower maintenance costs while supporting all major functions effectively.

2. The researchers recommend adding new features, such as AI-based chat support for inquiries, real-time email notifications for enrollment updates, and integration with external platforms. These enhancements would help the system meet a wider range of user needs and improve overall accessibility and responsiveness.

3. The researchers recommend enhancing the user interface of the online enrollment system to ensure it is intuitive and accessible for a wider range of users. Improving navigation, layout, and visual clarity can reduce user errors and create a smoother interaction for students,

parents, and faculty members. Incorporating responsive design and clearer prompts would further improve overall usability and user satisfaction.

4. The researchers recommend that the Online Enrollment System undergo regular updates to maintain its performance, security, and compatibility with new technologies. Consistent system improvements will prevent technical issues, support changing user needs, and ensure long-term reliability and sustainability.

5. The researchers recommend enhancing the system's data protection and privacy measures to safeguard sensitive student information. Strengthening these controls would reduce security risks and promote greater trust among users and administrators.

6. The researchers recommend optimizing the system's performance across different internet speeds and device types. Improving responsiveness and load handling will make the platform more accessible, especially for users with limited connectivity or older devices.

7. The researchers recommend incorporating a built-in reporting and issue-tracking feature to gather user concerns

directly within the system. This would support timely resolution of problems and guide future updates based on real user experiences and needs.

8. The researchers recommend integrating multilingual support to accommodate non-native English speakers. Providing language options would enhance accessibility and make the system more inclusive for a diverse user group.

9. The researchers recommend developing an analytic tool that tracks enrollment trends, user activity, and completion rates. These insights would support planning, reporting, and continuous system improvement.

10. The researchers recommend integrating a student portal feature that allows learners to securely view their grades and academic standing. Providing direct access to this information would improve transparency, reduce manual requests, and enhance the overall user experience for both students and administrators.